Kabul University
Computer Science Faculty
Software Engineering Department

Thesis/Final Project Proposal

Research/Project Title Afghanistan Innovation Management System

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1. Introduction

Innovation plays a central role in shaping the modern knowledge-based economy, driving progress in technology, business, and governance. It empowers countries to improve productivity, stimulate employment, and develop sustainable competitive advantages. According to (P. Vrgović & Jošanov-Vrgović, 2018) open innovation systems encourage collaboration and knowledge exchange among universities, industries, and entrepreneurs, allowing developing countries to overcome resource and infrastructure constraints. However, Afghanistan, as a developing nation, continues to face significant obstacles in fostering innovation due to the lack of an organized national framework and a centralized digital platform that supports innovators and policymakers (Hashimi & Azeem, 2025; P. Vrgović & Jošanov-Vrgović, 2018)).

In global practice, the ISO 56002 Innovation Management System (IMS) provides a standardized model that guides organizations in systematically managing their innovation processes. It defines principles such as leadership commitment, process structure, opportunity identification, idea evaluation, and implementation (Rezak et al., 2023). Implementing such a framework enhances organizational learning, promotes sustainability, and aligns innovation efforts with long-term development goals. For countries like Afghanistan, adopting an ISO-based system could enable consistency in managing innovation across institutions and create pathways for digital collaboration (Novillo-Villegas et al., 2022).

The absence of a digital innovation management framework in Afghanistan has limited coordination between innovators, universities, and government bodies. Many innovative ideas remain undocumented or fail to reach the stage of implementation due to weak communication, funding gaps, and lack of systematic evaluation. Therefore, this research proposes the design and development of a web-based Afghanistan Innovation Management System (AIMS), a digital platform that will centralize the processes of idea submission, evaluation, and progress tracking. The system will serve as a bridge connecting innovators, industries, and policymakers, ensuring transparency and inclusiveness (Ahmed et al., 2022,Saber Niazy, 2025).

Studies in other developing economies show that implementing digital innovation management platforms significantly enhances collaboration, efficiency, and innovation (Halimy et al., 2023). By following ISO 56002 standards, the proposed AIMS will help standardize innovation processes in Afghanistan, provide measurable results, and support national goals for economic transformation. Ultimately, AIMS, aims to move Afghanistan toward a sustainable and innovation-driven economy where ideas can grow into impactful projects contributing to social and economic development (Zhang et al., 2023)

2. Problem Statement

Innovation has been recognized globally as the foundation of sustainable economic growth and technological advancement. Yet, in Afghanistan, the innovation ecosystem remains underdeveloped and fragmented. Although universities and private organizations often generate innovative ideas, there is no structured mechanism to manage or evaluate them effectively. This lack of coordination results in duplication of efforts, missed opportunities, and a weak link between innovation, policy, and practice (Amani & Sarmashq, 2024; Mirza & Baharudin, 2024)

As stated by (P. Vrgović & Jošanov-Vrgović, 2018), effective innovation requires structured systems that facilitate collaboration, information sharing, and continuous improvement. However, Afghanistan lacks such a system. The innovation landscape is characterized by low digital literacy, inadequate infrastructure, limited research funding, and an absence of standardized

processses for idea management. In many cases, innovation initiatives are short-lived because they rely on informal or disconnected channels rather than an institutionalized framework (Halimy et al., 2023; Hashimi & Azeem, 2025).

According to Rezak et al., (2023), the ISO 56002 standard provides a practical model for innovation management, focusing on planning, evaluation, and systematic improvement. Without adopting similar standards, Afghan institutions face difficulties in sustaining innovative activeities and achieving measurable impact. The absence of an integrated national system also restricts communication between universities, industries, and government sectors leading to isolated innovation efforts with little scalability or replication (Kihlander et al., 2024; Novillo-Villegas et al., 2022).

Moreover, studies by Halimy et al., (2023) indicate that digital transformation in developing countries enhances organizational agility and innovation capability(Ahmed et al., 2022). Afghanistan, however, has not yet fully utilized digital tools to manage its innovation processes. There is currently no unified online platform to submit, evaluate, and monitor innovative ideas transparently. This creates inefficiencies and prevents policymakers from tracking progress and measuring outcomes (Zhang et al., 2023).

Therefore, the main problem addressed by this research is the absence of a centralized and standardized digital platform for innovation management in Afghanistan. This gap limits the country's capacity to promote collaboration, transparency, and accountability in its innovation ecosystem. The project aims to develop a web-based Afghanistan Innovation Management System (AIMS) aligned with ISO 56002 guidelines to ensure structured innovation, enhance communication among stakeholders, and promote evidence-based decision-making (Amani & Sarmashq, 2024; Rezak et al., 2023).

3. Significance of Research/Project

The Afghanistan Innovation Management System (AIMS) plays a crucial role in fostering innovation, collaboration, and transparency within Afghanistan's evolving digital economy. By connecting innovators, universities, industries, and policymakers on a unified online platform, AIMS supports Afghanistan's transition toward a knowledge-based and innovation-driven (Novillo-Villegas et al., 2022; Ziarahman Amani & Sarwanaaz Sarmashq, 2024).

The platform will enhance collaboration and accountability by facilitating open data sharing, idea evaluation, and progress monitoring, helping reduce bias and improve decision-making in innovation-related projects (P. Vrgović & Jošanov-Vrgović, 2018). Its digital and web-based architecture provides scalability and adaptability, ensuring that the system remains responsive to Afghanistan's future innovation and development needs (Halimy et al., 2023).

Furthermore, implementing AIMS in alignment with the ISO 56002 Innovation Management System standard ensures that innovation activities in Afghanistan follow global best practicespromoting efficiency, sustainability, and continuous (Kihlander et al., 2024) This alignment is especially important for building institutional capacity and encouraging cross-sectoral integration, which are vital for sustainable innovation ecosystems (Rezak et al., 2023).

Ultimately, AIMS contributes to overcoming Afghanistan's existing innovation barriers such as limited infrastructure and fragmented coordination by enabling inclusive participation and structured innovation management. This research will thus provide a model for how developing countries can leverage digital systems to support socio-economic growth, digital transformation, and long-term sustainability (Amani & Sarmashq, 2024; P. Vrgović & Jošanov-Vrgović, 2018).

4. **Research/Project Objectives**

4.1. Main Objective:

To design and implement a web-based Innovation Management System for Afghanistan (AIMS) using Spring Boot and React, enabling the structured submission, evaluation, and tracking of innovative ideas in alignment with international innovation standards (Kihlander et al., 2024; Rezak et al., 2023)).

4.2. **Sub Objectives:**

4.2.1. To develop a secure, inclusive, and user-friendly platform that connects innovateors, universities, industries, and policymakers, ensuring accessibility and effective collaboration across Afghanistan's innovation ecosystem (Amani & Sarmashq, 2024).

4.2.2. To create an evaluation and monitoring system that promotes transparency, accountttability, and data-driven decision-making for innovation projects reducing bias and ensuring fair assessment of submitted ideas (Kihlander et al., 2024)

4.2.3. To generate analytical reports and performance dashboards that assist decision-makers in tracking innovation outcomes, identifying key challenges, and supporting evidence-based policymaking for sustainable development (Halimy et al., 2023).

5. **Research Questions/ Problems which Project Solves**

5.1. Main Question/Problem:

How can a web-based innovation management system enhance innovation practices in Afghanistan by providing a structured, transparent, and collaborative digital platform for idea generation, evaluation, and implementation? (Ahmed et al., 2022).

5.2. Sub Questions/ Problem:

5.2.1. What core technological and functional components are essential for developing an effective and sustainable Innovation Management System aligned with Afghanistan's digital transformation goals? (Zhang et al., 2023)

5.2.2. How can digital tools and data-driven mechanisms improve the evaluation, monitoring, and reporting of innovation projects to ensure transparency, accountability, and evidence-based decision-making? (Rezak et al., 2023)

5.2.3. How can the system ensure inclusiveness and accessibility for diverse Afghan innovators such as students, entrepreneurs, and researchers through user-centered and collaborative design approaches? (Mirza & Baharudin, 2024).

6. **Literature Review**

Innovation management systems (IMS) are increasingly recognized as essential mechanisms for driving economic growth, competitiveness, and organizational sustainability. Effective innovation management involves structured processes that enable organizations to generate, evaluate, and implement innovative ideas in a systematic manner (Rezak et al., 2023)). The ISO 56002 Innovation Management System standard provides a globally accepted framework outlining the principles, structure, and tools necessary for managing innovation activities strategically- (Kihlander et al., 2024).

Research confirms that adopting ISO 56002 enhances organizations' ability to maintain continuous improvement, gain competitive advantage, and strengthen their innovation culture (Rezak et al., 2023). These standards also ensure that innovation is managed in alignment with long-term sustainability and digital transformation (Rezak et al., 2023,P. V. & I. J. Vrgović, 2018).

Digital transformation plays a crucial role in supporting innovation systems. Studies have shown that digital innovation platforms enable collaboration, transparency, and efficient knowledge sharing among different stakeholders (P. Vrgović & Jošanov-Vrgović, 2018). In developing

coun-tries, digital systems have been proven to overcome traditional barriers such as limited infrastructure, fragmented coordination, and lack of innovation awareness (Halimy et al., 2023). These platforms also empower small and medium enterprises (SMEs) by facilitating access to global networks and new markets through sustainable digital collaboration (Hashimi & Azeem, 2025).

In the Afghan context, innovation remains constrained by weak institutional frameworks, insufficient digital infrastructure, and limited collaboration between academia, industry, and government. According to Ziarahman Amani & Sarwanaaz Sarmashq, (2024); emphasize the need for public universities and research institutions in Afghanistan to play a more active role in fostering innovation, creativity, and employment opportunities. Similarly, Halimy et al., (2023); highlight that digital transformation initiatives can enhance Afghanistan's adaptive capability and competitiveness if implemented through structured and scalable platforms.

The successful design of innovation management systems requires attention to critical factors such as leadership commitment, organizational culture, and system integration across depart-ents (Kihlander et al., 2024,Samuel, 2014). Integrating these principles within Afghanistan's socioeconomic context can help establish a robust innovation ecosystem supported by evidence-based decision-making and cross-sector collaboration.

From a technological standpoint, adopting modern web frameworks such as Spring Boot (for backend) and React (for frontend) ensures that the proposed Afghanistan Innovation Manage-ment System (AIMS) will be secure, scalable, and user-friendly. These technologies support modular design, efficient data handling, and smooth interaction among users, making them suitable for managing complex innovation workflows (Halimy et al., 2023).

In summary, prior studies collectively demonstrate that implementing an Innovation Management System based on ISO 56002, supported by digital collaboration platforms, can significantly improve innovation outcomes. For Afghanistan, establishing AIMS offers a transformative opportunity to integrate global best practices with local innovation needs thereby contributing to national goals of digital transformation, knowledge-based development, and sustainable economic growth (Novillo-Villegas et al., 2022).

7. Research/Project Methodology

This research employs a literature-based and design-oriented methodology to develop a concept-tual model for the Afghanistan Innovation Management System (AIMS). Since access to primary data is limited due to institutional and infrastructural challenges in Afghanistan, this approach relies on credible secondary data sources, including international journals, ISO standards, and case studies from developing economies. The methodology ensures that all design decisions are grounded in globally recognized practices while being adaptable to Afghanistan's local context.

7.1. Research Design:

The study follows a qualitative, descriptive, and conceptual research design. The focus is on understanding the key principles of innovation management systems and how they can be adapted to Afghanistan's socio-economic environment. The design integrates theore-tical insights from previous studies (e.g., (Rezak et al., 2023)) with practical design considerations for web-based systems.

7.2. **Data Collection:**

Data are collected entirely from secondary sources, including open-access journals, conference papers, and ISO documentation. The selected sources provide evidence on innovation management models, success factors, and challenges in implementing digital innovation systems. According to Major references include Vrgović & Jošanov-Vrgović, (2018)for open innovation theory, Rezak et al., (2023) for ISO 56002 structure, and Halimy et al., (2023) for digital transformation insights in developing countries. These studies collectively offer the theoretical basis for designing AIMS.

7.3. System Design:

Based on the analyzed data, the project will propose a conceptual web-based platform architect-ure developed using Spring Boot for the backend and React for the frontend. The design will include key modules such as:

- Idea Submission and Evaluation
- Project Monitoring and Reporting
- Data Analytics and Dashboards
- User Roles and Access Management

This architecture aligns with the ISO 56002 framework and ensures scalability, security, and inclusiveness for Afghan institutions.

7.4. Justification:

The literature-based approach is both feasible and suitable for Afghanistan's current research environment. It minimizes logistical barriers and ensures that the project remains grounded (Vrgović & Jošanov-Vrgović, 2018), proven international models. This methodology provides a strong academic foundation while allowing flexibility for future implementation. The combination of literature synthesis and conceptual design ensures that AIMS will be both academically credible and practically viable for Afghanistan's innovation development.

8. Expected Outcomes

The implementation of the Afghanistan Innovation Management System (AIMS) is expected to produce several key outcomes that advance the nation's innovation capacity. First, it will establish a centralized and transparent platform for idea submission, evaluation, and monitoring, fostering collaboration among universities, industries, and policymakers (Ziarahman Amani & Sarwanaaz Sarmashq, 2024). Second, by adopting principles from ISO 56002, AIMS will introduce a structured and standardized framework for managing innovation that promotes accountability, efficiency, and continuous Rezak et al., (2023) The digital architecture developed using Spring Boot and React will ensure scalability, security, and accessibility for diverse users across Afghanistan (Halimy et al., 2023). Additionally, the system will generate data-driven analytical reports to support evidence-based decision-making and policy formulation. Overall, AIMS will strengthen Afghanistan's national innovation ecosystem, enhance knowledge sharing, and contribute to the country's transition toward a sustainable, knowledge-based digital economy (P. Vrgović et al., 2024).

9. Work Plan

Semester7:

Activities	Weeks									
	1	2	3	4	5	6	7	8	9	10
Topic selection										
Submitting proposal										
Literature Review										
Problem and Objectives Definition										
Work report										
Report review (by committee)										

Semester8:

Activities	Weeks															
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Methodology																
Project Design																
Implementation/analysis																
Result																
Pre Defense																
Evaluation																
Conclusion/ Outcome																
Project Report																

10) References

[1]. Ahmed, R. K., Ahmed, O., Pappel, I., Reitsakas, A., & Draheim, D. (2022). The Role of Digital Transformation in Fostering Transparency: An e-Court System Case Study. *Lecture Notes in Computer Science (Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics), 13454 LNCS*, 219–230. https://doi.org/10.1007/978-3-031-15342-6_17

[2]. Amani, Z., & Sarmashq, S. (2024). *International Journal of Asian and African Studies Exploring Innovation, Creativity, and Employment in Public Universities of Afghanistan*. https://doi.org/10.32996/ijaas

[3]. Halimy, M. S., Liu, J., & Tawheed, Q. (2023). Research on Digital Transformation Innovation and Adaptive Capability of Manufacturing Enterprises in Afghanistan. *OALib*, *10*(08), 1–17. https://doi.org/10.4236/oalib.1110498

[4]. Hashimi, M. A., & Azeem, M. (2025). Barriers to digital transformation in SMEs: A case study of Afghanistan. *Asian Journal of Management and Commerce*, *6*(1), 829–833. https://doi.org/10.22271/27084515.2025.v6.i1i.536

[5]. Kihlander, I., Magnusson, M., & Karlsson, M. (2024). Critical Factors to Consider When Designing an Innovation Management System. *Research Technology Management*, *67*(3), 34–43. https://doi.org/10.1080/08956308.2024.2323893

[6]. Mirza, A., & Baharudin, S. 'Aisyah. (2024). INNOVATION BARRIER TO SMES DEVELOPMENT: EVIDENCE FROM AFGHANISTAN. *International Journal of Business and Society*, *25*(1), 388–406. https://doi.org/10.33736/ijbs.6922.2024

[7]. Novillo-Villegas, S., Ayala-Andrade, R., Lopez-Cox, J. P., Salazar-Oyaneder, J., & Acosta-Vargas, P. (2022). A Roadmap for Innovation Capacity in Developing Countries. *Sustainability (Switzerland)*, *14*(11). https://doi.org/10.3390/su14116686

[8]. Rezak, R., Djenouhat, A., & Kherbachi, H. (2023). The Role of Innovation Management System ISO 56002 Standard in the Competitive Edge toward Sustainability. *Sumerianz Journal of Business Management and Marketing*, *61*, 1–8. https://doi.org/10.47752/sjbmm.61.1.8

[9]. Saber Niazy, M. (2025). *The Role of Technology in Enhancing Small Business Operations in Afghanistan*. www.universepg.com

[10]. Samuel, P. (2014). No Title. *International Journal of Technology Management and Sustainable Development*, *13*(3), 201–215.

[11]. Vrgović, P., & Jošanov-Vrgović, I. (2018). Open innovation systems in developing countries: Sustainable digital networks and collaboration in SMEs. *Dynamic Relationships Management Journal*, *7*(2), 5–11. https://doi.org/10.17708/DRMJ.2018.v07n02a01

[12]. Vrgović, P., Jošanov-Vrgović, I., Ziarahman Amani, Sarwanaaz Sarmashq, Rezak, R., Djenouhat, A., Kherbachi, H., Kihlander, I., Magnusson, M., Karlsson, M., Halimy, M. S., Liu, J., Tawheed, Q., Rad, O., Hashimi, M. A., Azeem, M., Kihlander, I., Magnusson, M., & Karlsson, M. (2024). Critical Factors to Consider When Designing an Innovation Management System. *Research Technology Management*, *67*(3), 34–43. https://doi.org/10.1080/08956308.2024.2323893

[13]. Vrgović, P. V. & I. J. (2018). No Title. *Dynamic Relationships Management Journal (DRMJ)*, *7*(2 (November 2018)), 5–11. https://www.sam-d.si/wp-content/uploads/2018/12/DRMJ.2018.v07n02a01-1.pdf

[14]. Zhang, J., van Gorp, D., & Kievit, H. (2023). Digital technology and national entrepreneurship: An ecosystem perspective. *Journal of Technology Transfer*, *48*(3), 1077–1105. https://doi.org/10.1007/s10961-022-09934-0

[15]. Ziarahman Amani, & Sarwanaaz Sarmashq. (2024). Exploring Innovation, Creativity, and Employment in Public Universities of Afghanistan. *International Journal of Asian and African Studies*, *3*(1), 80–85. https://doi.org/10.32996/ijaas.2024.3.1.9

Date and Signature of Supervisor and Student

Supervisor and student	Name and Last name	Signature	Date
Supervisor			
Student	Farid Gul Shakir		
Student	Sayedagha Hotak		

الف) تایید استاد رهنما

موضوع و طرح تحقیق محترم ..... تحت عنوان.....از طرف استاد رهنما محترم..... مورخ ..... تایید است

امضا استاد رهنما

ب) تایید کمیته دیپارتمنت

از نظر کمیته بررسی مونوگراف دیپارتمنت .....، پیشنهاد تعیین موضوع تحقیقی یا پروژه محترم ..... به شرح فوق را در جلسه مورخ .....به اتفاق آرا تایید است.

امضا اعضای کمیته

د) تایید دیپارتمنت

از نظر دیپارتمنت..... در جلسه مورخ ..... پیشنهاد تعیین موضوعی تحقیقی یا پروژه محترم /محترمه ..... را به اتفاق آرا تایید است.

با احترام

پوهنیار سلیم همدرد
سرپرست دیپارتمنت انجینری نرم افزار